T04 (A3): Contraction without beta-min for the SMC-BVS v3 target

Proof, version 2 (after review round 1)

Wave 3, lead Claude Opus 5.5. Status: under review; not adopted.

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The statement is that of `STATEMENT.md`, which is restated in Section 0. Everything is proved from the definitions there (including the v3 definition of π, quoted in §0), the assumptions G, R, SE, DX, and the single hypothesis H-T02 (= T02(i)). No other result is used, apart from elementary facts that are proved inline.

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0 Setting, hypotheses and the claim

Data and contrast coordinates. Throughout, $n \geq 4$, $p \geq 1$ and $N := n - 1$. The raw design $X^r \in \mathbb{R}^{n\times p}$ is fixed and has no constant column. $C = I_n - \mathbf{1}\mathbf{1}^\top/n$, D_x is the diagonal matrix of raw column sample standard deviations, and $X = CX^rD_x^{-1}$ is the standardised design, so that $\|x_j\|^2 = N$ for every column. $U \in \mathbb{R}^{n\times N}$ is a fixed matrix with $U^\top U = I_N$ and $UU^\top = C$. We write

$$Z := U^\top X \in \mathbb{R}^{N\times p} \quad (\text{so } X = UZ), \qquad y_c := U^\top y^r \in \mathbb{R}^N, \qquad s_y^2 := \|Cy^r\|^2/N = \|y_c\|^2/N.$$

Z_S denotes the columns of Z indexed by $S \subseteq [p]$, and $|S| = k$. Since $X = UZ$ and $U^\top U = I$,

$$G_S := X_S^\top X_S = Z_S^\top Z_S, \qquad \|z_j\|^2 = \|x_j\|^2 = N, \qquad R_S := G_S/N.$$

Further $H_S := (n+1)G_S + I_k$, $D_S := \det H_S / \det(G_S + I_k)$, $M_S := nZ_S H_S^{-1} Z_S^\top \in \mathbb{R}^{N\times N}$, and P_S is the orthogonal projection of $\mathbb{R}^N$ onto the column space of Z_S. For $S = \emptyset$: $G_\emptyset$ is the empty matrix, $D_\emptyset = 1$, and $M_\emptyset = P_\emptyset = 0$.

Truth. $y^r = \alpha^* \mathbf{1} + X^r \beta^* + \varepsilon$ with $\varepsilon \sim N_n(0, \sigma_\star^2 I_n)$, $\sigma_\star^2 > 0$ fixed. $\mu = CX^r\beta^* = X\theta^*$ with $\theta^* = D_x \beta^*$, $S_0 = \operatorname{supp}\beta^* = \operatorname{supp}\theta^*$ and $s_0 = |S_0|$. $\mathbb{P}_0, \mathbb{E}_0$ refer to the law of ε. We put

$$\mu_c := U^\top \mu = Z\theta^*, \qquad e := U^\top \varepsilon, \qquad E := \|e\|^2/\sigma_\star^2.$$

Since $U^\top \mathbf{1} = U^\top C \mathbf{1} = 0$ and $U^\top X^r = U^\top C X^r$, we have $y_c = \mu_c + e$. Since $U^\top U = I_N$, $e \sim N_N(0, \sigma_\star^2 I_N)$ and $E \sim \chi^2_N$. Also $\mu \in \operatorname{range} C$, so $\mu = U\mu_c$ and $\|\mu\| = \|\mu_c\|$.

Prior and target. $\Omega = \{S \subseteq [p] : |S| \le K\}$ with $K = \min(p, \lfloor n/2 \rfloor)$. $w(k) = \frac{1}{2}/\binom{2p-1}{k}$, $W_K = \sum_{k\le K}\binom{p}{k} w(k)$ and $\varpi(S) = w(|S|)/W_K$ on Ω. $b_0 = 0.01$ and $\nu = 0.01 + n/2$. π is the v3 posterior on Ω, defined in 00_COMMON §1 (and in `STATEMENT.md`) as follows: with $y := Cy^r/s_y$, $z_S := X_S^\top y$, $Q_S^{\mathrm{v3}} := y^\top y - n z_S^\top H_S^{-1} z_S$ and $a_0 = b_0 = 0.01$,

$$m(S) = \frac{(2b_0)^{a_0}\Gamma(\nu)}{\pi^{n/2}\Gamma(a_0)} D_S^{-1/2}\big(2b_0 + Q_S^{\mathrm{v3}}\big)^{-\nu}, \qquad \pi(S) = \frac{\varpi(S)m(S)}{\sum_{S'\in\Omega}\varpi(S')m(S')} \quad (S \in \Omega).$$

(In $m(S)$, $\pi^{n/2}$ is a power of the number π.) $\Pi(\cdot \mid y)$ denotes π as a law on S.

The joint contrast posterior $\Pi^\dagger$. Put $b := b_0 s_y^2$. $\Pi^\dagger(\cdot \mid y)$ is the posterior of (S, θ, v), with $\theta \in \mathbb{R}^p$ and $\theta_j = 0$ for $j \notin S$, in the model

$$S \sim \varpi, \quad v \sim \mathrm{IG}(0.51, b), \quad \theta_S \mid v, S \sim N_{|S|}\big(0, nv(G_S + I)^{-1}\big), \quad y_c \mid \theta, v, S \sim N_N(Z_S\theta_S, vI_N).$$

Here $\mathrm{IG}(a, c)$ has density $\propto v^{-a-1}e^{-c/v}$ on $(0,\infty)$. The model is well defined when $b > 0$, that is, when $y_c \neq 0$. This holds $\mathbb{P}_0$-almost surely, because y_c is Gaussian with covariance $\sigma_\star^2 I_N$. All statements below are made on $\{y_c \neq 0\}$. Since $X\theta = UZ\theta$ and $\mu = U\mu_c$,

$$\|X\theta - \mu\| = \|Z_S\theta_S - \mu_c\| \qquad \text{for every } S \text{ and every } \theta \text{ supported in } S. \tag{1}$$

Rates. $\ell_n := \log(p \vee n)$, $\varepsilon_{2,n}^2 := (s_0+1)\ell_n^2/n$, $\varepsilon_{1,n}^2 := (s_0+1)\ell_n/n$, $d_* := s_0 + \lceil s_0(2A+1)\rceil$, and

$$m_n := \Big\lceil \frac{s_0\log(2p) + \frac{s_0+1}{2}\log n + \frac{1}{2}\log n}{\log 2} \Big\rceil.$$

Note that $\ell_n \ge \log 4 > 1$, $\log p \le \ell_n$, $\log 2 \le \ell_n$ and $\log(n+1) \le 2\log n \le 2\ell_n$.

Assumptions.

R1 $n \to \infty$, s_0 fixed, $p = p_n \le n^A$ for a fixed $A \ge 1$. Hence $\ell_n \le A \log n$.

G The fixed design and Gaussian errors above.

R s_0 fixed; $p \le n^A$; $\|\mu\|^2/n \le M$; $\lambda_{\min}(R_{S_0}) \ge \varphi_0^2 > 0$ (vacuous if $s_0 = 0$); $s_0 \le K$ for all large n.

SE (where stated) $\lambda_{\min}(R_T) \ge \varphi_0^2$ for every nonempty $T \subseteq [p]$ with $|T| \le \min(p, m_n + s_0)$.

DX (for (iv)) every diagonal entry of D_x is at least $d_- > 0$.

H-T02 (T02(i), taken as a hypothesis) In R1 under G and R, $\Pi(|S| > m_n \mid y) \to 0$ in $\mathbb{P}_0$-probability. (T02(i) states $\Pi(|S| > m_n(\omega_n) \mid y) = O_{\mathbb{P}_0}(e^{-\omega_n})$ for every deterministic $\omega_n \to \infty$ with $\omega_n = O(\log n)$. We use it only with $\omega_n = \frac{1}{2}\log n$, for which $m_n(\omega_n) = m_n$.)

The claim (Theorem T04). "$\to 0$" means convergence to 0 in $\mathbb{P}_0$-probability. There is a constant $C < \infty$, depending only on $(\sigma_\star^2, M, \varphi_0, A)$, such that in R1 under G, R and H-T02:

(i) $\Pi^\dagger(\|X\theta - \mu\|^2/n > C\varepsilon_{2,n}^2 \mid y) \to 0$;

(ii) under SE in addition: $\Pi(|S| > d_* \mid y) \to 0$; $\Pi^\dagger(\|X\theta - \mu\|^2/n > C\varepsilon_{1,n}^2 \mid y) \to 0$; and $\Pi^\dagger(\|\theta - \theta^*\|^2 > C\varepsilon_{1,n}^2/\varphi_0^2 \mid y) \to 0$;

(iii) for every deterministic $u_n \to \infty$: $\Pi^\dagger(|v/\sigma_\star^2 - 1| > C[\varepsilon_n^2 + u_n/\sqrt{n}] \mid y) \to 0$, with $\varepsilon_n = \varepsilon_{2,n}$ under G, R, and $\varepsilon_n = \varepsilon_{1,n}$ under SE in addition;

(iv) under SE and DX: $\Pi^\dagger(\|D_x^{-1}(\theta - \theta^*)\|^2 > C\varepsilon_{1,n}^2/(\varphi_0^2 d_-^2) \mid y) \to 0$.

The constant C is given explicitly in (12).

Notation. Some letters carry fixed meanings that differ from nearby symbols: C is the centring matrix, while the constant of the theorem is C in (12) (no confusion arises, since the centring matrix appears only in §0–§1); M is the signal bound of R, while M_S is a matrix and $\mathcal{M}$ a set of models; B is a constant, $\mathcal{B}(d)$ a set, and $\mathrm{B}(\cdot,\cdot)$ the Beta function; W_K is the prior normaliser and W_S a noise term; v is the variance parameter and v_S a squared norm; E is a χ^2_N variable and $\mathbb{E}$ an expectation; $\Gamma(\cdot)$ is the Gamma function.

Plan. Section 1 computes $\Pi^\dagger$ and shows that its S-marginal is π. Section 2 collects the matrix algebra. Section 3 defines a high-probability noise event. Section 4 shows that models with a large ridge approximation error have negligible posterior mass. Section 5 turns this into a contraction bound for any dimension cut-off d (the master theorem). Section 6 takes $d = m_n$ (from H-T02) and proves (i) and the first case of (iii). Section 7 proves the sharp dimension bound $\Pi(|S| > d_* \mid y) \to 0$ under SE. Section 8 takes $d = d_*$ and proves (ii), the second case of (iii), and (iv).

1 The contrast posterior and the prior

For $S \subseteq [p]$ put

$$Q_S := y_c^\top (I_N - M_S) y_c.$$

Lemma 1.1 (The contrast posterior)**.** *On $\{y_c \neq 0\}$, for every $S \in \Omega$:*

(a) $\Pi^\dagger(S \mid y) = \varpi(S) D_S^{-1/2} (2b + Q_S)^{-\nu} \big/ \sum_{S' \in \Omega} \varpi(S') D_{S'}^{-1/2} (2b + Q_{S'})^{-\nu}$;

(b) given (S, y), $v \sim \mathrm{IG}(\nu,\, b + Q_S/2)$;

(c) given (v, S, y), $\theta_S \sim N_{|S|}\big(nH_S^{-1}Z_S^\top y_c,\ nvH_S^{-1}\big)$.

Proof. Fix S with $k = |S|$ and write $\Sigma_S := I_N + nZ_S(G_S + I)^{-1}Z_S^\top$. Integrating θ_S out of the Gaussian model gives $y_c \mid v, S \sim N_N(0, v\Sigma_S)$. By Sylvester's determinant identity,

$$\det \Sigma_S = \det\big(I_k + n(G_S + I)^{-1}G_S\big) = \det\big((G_S + I)^{-1}((n+1)G_S + I)\big) = D_S.$$

By the Woodbury identity with $A = n(G_S + I)^{-1}$, so that $(A^{-1} + Z_S^\top Z_S)^{-1} = \big(((n+1)G_S + I)/n\big)^{-1} = nH_S^{-1}$, we get $\Sigma_S^{-1} = I_N - nZ_SH_S^{-1}Z_S^\top = I_N - M_S$. Hence the density of y_c given (v, S) is $(2\pi v)^{-N/2}D_S^{-1/2}\exp(-Q_S/(2v))$. Multiplying by the $\mathrm{IG}(0.51, b)$ density $b^{0.51}\Gamma(0.51)^{-1}v^{-1.51}e^{-b/v}$, the joint density of (y_c, v) given S is

$$(2\pi)^{-N/2}D_S^{-1/2}\frac{b^{0.51}}{\Gamma(0.51)}\,v^{-(0.51+N/2)-1}\exp\Big(-\frac{b + Q_S/2}{v}\Big).$$

Note that $0.51 + N/2 = 0.01 + n/2 = \nu$. As a function of v this is an $\mathrm{IG}(\nu, b + Q_S/2)$ kernel, which gives (b). Integrating over v gives the marginal density

$$(2\pi)^{-N/2}\frac{b^{0.51}\Gamma(\nu)}{\Gamma(0.51)}\,D_S^{-1/2}\,(b + Q_S/2)^{-\nu}.$$

Its S-dependent part is $D_S^{-1/2}(2b + Q_S)^{-\nu}$, up to the S-free factor 2^ν. Multiplying by $\varpi(S)$ and normalising gives (a). Here $2b + Q_S \geq 2b > 0$, because $I - M_S \succeq 0$ (Lemma 2.1(a), which does not depend on this lemma). For (c): given (v, S) the log posterior density of θ_S is, up to constants, $-\frac{1}{2v}\big(\|y_c - Z_S\theta_S\|^2 + \theta_S^\top(G_S + I)\theta_S/n\big)$. This is a Gaussian kernel with precision $(G_S + (G_S + I)/n)/v = H_S/(nv)$ and mean $(H_S/(nv))^{-1}Z_S^\top y_c/v = nH_S^{-1}Z_S^\top y_c$. □

Lemma 1.2 (The S-marginal of $\Pi^\dagger$ is π)**.** *On $\{y_c \neq 0\}$, $\Pi^\dagger(S \mid y) = \pi(S)$ for every $S \in \Omega$.*

Proof. By the definition in §0, $\pi(S) \propto \varpi(S)D_S^{-1/2}(2b_0 + Q_S^{\mathrm{v}3})^{-\nu}$, with $Q_S^{\mathrm{v}3} = y^\top y - nz_S^\top H_S^{-1}z_S$, $y = Cy^r/s_y$ and $z_S = X_S^\top y$ (the factor in front of $D_S^{-1/2}$ does not depend on S). Since $Cy^r = UU^\top y^r = Uy_c$, we have $y = Uy_c/s_y$. Hence $y^\top y = \|y_c\|^2/s_y^2$ and $z_S = Z_S^\top U^\top Uy_c/s_y = Z_S^\top y_c/s_y$. Therefore $Q_S^{\mathrm{v}3} = Q_S/s_y^2$ and $2b_0 + Q_S^{\mathrm{v}3} = (2b + Q_S)/s_y^2$. The factor $s_y^{2\nu}$ does not depend on S, so the normalised weights agree with Lemma 1.1(a). (D_S is the same object in both, since $G_S = X_S^\top X_S$.) □

From now on $\Pi(\cdot \mid y)$ denotes both π and the S-marginal of $\Pi^\dagger$. For $\mathcal{A} \subseteq \Omega$ we write $\Pi(\mathcal{A} \mid y) = \sum_{S\in\mathcal{A}}\pi(S)$, and

$$\frac{\pi(S)}{\pi(S')} = \frac{\varpi(S)}{\varpi(S')}\Big(\frac{D_{S'}}{D_S}\Big)^{1/2}\Big(\frac{2b + Q_{S'}}{2b + Q_S}\Big)^{\nu} \qquad (S, S' \in \Omega). \tag{2}$$

Lemma 1.3 (The prior)**.** *(a) $\sum_{k=0}^{p}\binom{p}{k}w(k) = 1$; hence $W_K \leq 1$ and $\sum_{S\in\Omega}\varpi(S) = 1$.*

(b) $w(k)/w(k+1) = (2p - 1 - k)/(k+1) \leq 2p$ for $0 \leq k \leq p - 1$.

(c) If $s_0 \leq K$, then $\varpi(S_0) \geq w(s_0) \geq \frac{1}{2}(2p)^{-s_0}$.

(d) If $j \geq 1$ and $s_0 + j \leq p$, then $w(s_0 + j)/w(s_0) \leq \big((s_0 + j)/p\big)^j$.

Proof. (a) With the Beta function, $p\,\mathrm{B}(k+1,2p-k) = p\,k!(2p-1-k)!/(2p)! = \frac{1}{2}\,k!(2p-1-k)!/(2p-1)! = w(k)$. Since $\mathrm{B}(k+1,2p-k) = \int_0^1 t^k(1-t)^{p-k}(1-t)^{p-1}dt$, the binomial theorem gives

$$\sum_{k=0}^{p}\binom{p}{k}w(k) = p\int_0^1(1-t)^{p-1}\sum_{k=0}^{p}\binom{p}{k}t^k(1-t)^{p-k}\,dt = p\int_0^1(1-t)^{p-1}dt = 1.$$

So $W_K \le 1$, and $\sum_\Omega \varpi = W_K/W_K = 1$. (b) $\binom{2p-1}{k+1}/\binom{2p-1}{k} = (2p-1-k)/(k+1)$. (c) $W_K \le 1$ gives $\varpi(S_0) \ge w(s_0)$, and $\binom{2p-1}{s_0} \le (2p-1)^{s_0} \le (2p)^{s_0}$. (d) By (b), $w(s_0+j)/w(s_0) = \prod_{i=s_0}^{s_0+j-1}\frac{i+1}{2p-1-i}$. Each factor is at most the last one, $(s_0+j)/(2p-s_0-j)$, and $2p-s_0-j \ge p$. □

2 Matrix algebra

Define, for every $S \subseteq [p]$,

$$a_S := \mu_c^\top(I-M_S)\mu_c, \qquad W_S := \mu_c^\top(I-M_S)e, \qquad v_S := \|(I-M_S)\mu_c\|^2, \qquad \delta := \frac{\|\mu\|^2+\|\theta^*\|^2}{n}.$$

Lemma 2.1. *For every $S \subseteq [p]$ with $k = |S|$:*

(a) M_S is symmetric, $0 \preceq M_S \preceq \frac{n}{n+1}P_S \preceq P_S$, $M_S^2 \preceq M_S$, $\operatorname{tr} M_S \le k$, and $0 \preceq I - M_S \preceq I$ with $(I-M_S)^2 \preceq I - M_S$.

(b) For every $w \in \mathbb{R}^N$, $w^\top(I-M_S)w = \min_{t\in\mathbb{R}^k}\{\|w - Z_S t\|^2 + t^\top(G_S+I)t/n\}$. Consequently $0 \le Q_S \le \|y_c\|^2$, $0 \le v_S \le a_S \le \|\mu\|^2$, and $a_S \le \delta$ for every $S \supseteq S_0$.

(c) $Q_S = a_S + 2W_S + \sigma_\star^2 E - e^\top M_S e$, with $0 \le e^\top M_S e \le \|P_S e\|^2$. Hence $|Q_S - \sigma_\star^2 E| \le a_S + 2|W_S| + \|P_S e\|^2$.

(d) $1 \le D_S \le (n+1)^k$.

(e) $\|M_S y_c - \mu_c\|^2 \le 2a_S + 2\|P_S e\|^2$.

(f) If $S \subseteq S'$, then $Q_{S'} \le Q_S$ and $1 \le D_{S'}/D_S \le (n+1)^{|S'\setminus S|}$.

(g) If $k \ge 1$ and $\lambda_{\min}(G_S) \ge N\varphi_0^2 > 0$, then $M_S \succeq \big(1 - \frac{1+\varphi_0^{-2}}{n}\big)P_S$.

(h) If $S \subseteq S'$, $J := S' \setminus S$ has $j \ge 1$ elements, $\lambda_{\min}(G_{S'}) \ge N\varphi_0^2 > 0$, and $\|z_i\|^2 = N$ for $i \in J$, then $D_{S'}/D_S \ge \big(n\varphi_0^2/(j+1)\big)^j$.

(i) Under R, $\delta \le \bar\delta := M(1+\varphi_0^{-2})$.

Proof. (a) Let $G_S = V\,\mathrm{diag}(\lambda_1,\dots,\lambda_k)V^\top$ with V orthogonal and $\lambda_i \ge 0$. The vectors $Z_S V e_i$ are orthogonal with squared norms λ_i. For $\lambda_i > 0$ put $q_i := Z_S V e_i/\sqrt{\lambda_i}$; these form an orthonormal basis of the column space of Z_S. Then

$$M_S = \sum_{i:\lambda_i>0}\frac{n\lambda_i}{(n+1)\lambda_i+1}q_i q_i^\top, \qquad P_S = \sum_{i:\lambda_i>0} q_i q_i^\top,$$

and $0 < \frac{n\lambda}{(n+1)\lambda+1} < \frac{n}{n+1}$ for $\lambda > 0$. All claims follow.

(b) For $k = 0$ both sides equal $\|w\|^2$. For $k \ge 1$, the map $t \mapsto \|w - Z_S t\|^2 + t^\top(G_S+I)t/n$ is a strictly convex quadratic with Hessian $2H_S/n$. Its minimum is $\|w\|^2 - w^\top Z_S(H_S/n)^{-1}Z_S^\top w =$

$w^\top(I - M_S)w$. With $w = y_c$ this gives $Q_S \geq 0$, and with $t = 0$ it gives $Q_S \leq \|y_c\|^2$. With $w = \mu_c$, $t = 0$ gives $a_S \leq \|\mu_c\|^2 = \|\mu\|^2$. By (a), $v_S = \mu_c^\top (I - M_S)^2 \mu_c \leq a_S$. If $S \supseteq S_0$, take $t = \theta^*_S$ (zero off S_0). Then $Z_S t = \mu_c$ and $t^\top(G_S + I)t/n = (\|Z_S t\|^2 + \|t\|^2)/n = \delta$.

(c) Expand $(\mu_c + e)^\top (I - M_S)(\mu_c + e)$ and use (a).

(d) $D_S = \prod_i \frac{(n+1)\lambda_i + 1}{\lambda_i + 1}$, and each factor lies in $[1, n+1]$.

(e) $M_S y_c - \mu_c = -(I - M_S)\mu_c + M_S e$, so the left side is at most $2v_S + 2e^\top M_S^2 e \leq 2a_S + 2\|P_S e\|^2$, by (a) and (b).

(f) Let t attain the minimum in (b) for $(S, w = y_c)$. Let $t' \in \mathbb{R}^{|S'|}$ equal t on S and 0 on $S' \setminus S$. Then $Z_{S'} t' = Z_S t$ and $t'^\top (G_{S'} + I)t' = \|Z_{S'} t'\|^2 + \|t'\|^2 = t^\top (G_S + I)t$, so $Q_{S'} \leq Q_S$. For the determinants, order S' as (S, J) with $J = S' \setminus S$, $j = |J|$, and write $G_{S'}$ in blocks. By the Schur determinant formula,

$$\frac{D_{S'}}{D_S} = \frac{\det\big((n+1)R_2 + I_j\big)}{\det(R_1 + I_j)}, \qquad \begin{aligned} R_2 &:= G_{JJ} - (n+1)G_{JS}\big((n+1)G_S + I\big)^{-1} G_{SJ}, \\ R_1 &:= G_{JJ} - G_{JS}(G_S + I)^{-1} G_{SJ}. \end{aligned} \tag{3}$$

Indeed, the Schur complement of $(n+1)G_S + I$ in $(n+1)G_{S'} + I$ is $(n+1)R_2 + I$, and that of $G_S + I$ in $G_{S'} + I$ is $R_1 + I$. (For $S = \emptyset$, read $R_1 = R_2 = G_{JJ}$.) Both are positive definite, being Schur complements of positive definite matrices. $R_1 \succeq 0$, because it is the Schur complement of the positive definite block $G_S + I$ in the positive semidefinite matrix $G_{S'} + \mathrm{diag}(I_{|S|}, 0)$. Since $(n+1)((n+1)G_S + I)^{-1} = (G_S + \frac{1}{n+1} I)^{-1} \succeq (G_S + I)^{-1}$, we have $R_2 \preceq R_1$. Then $(n+1)R_2 + I \preceq (n+1)R_1 + I \preceq (n+1)(R_1 + I)$, and monotonicity of the determinant on positive definite matrices gives $D_{S'}/D_S \leq (n+1)^j$. For the lower bound $D_{S'}/D_S \geq 1$: write $D_{S'} = \prod_i \frac{(n+1)\lambda'_i + 1}{\lambda'_i + 1}$ as in (d). Then $\log D_T = \sum_i g(\lambda_i(G_T))$ with $g(\lambda) = \log \frac{(n+1)\lambda + 1}{\lambda + 1}$, which is nondecreasing on $[0, \infty)$ and has $g(0) = 0$. By Cauchy interlacing, the i-th largest eigenvalue of $G_{S'}$ is at least the i-th largest eigenvalue of its principal submatrix G_S ($i \leq |S|$), and the remaining eigenvalues of $G_{S'}$ are ≥ 0. Hence $\log D_{S'} \geq \log D_S$.

(g) In the notation of (a), every $\lambda_i \geq N\varphi_0^2 > 0$, so the q_i run over all i. For $\lambda \geq N\varphi_0^2$,

$$\frac{n\lambda}{(n+1)\lambda + 1} = \frac{n}{n+1} \cdot \frac{1}{1 + \frac{1}{(n+1)\lambda}} \geq \frac{n}{n+1}\Big(1 - \frac{1}{(n+1)\lambda}\Big) \geq 1 - \frac{1}{n+1} - \frac{1}{(n+1)N\varphi_0^2} \geq 1 - \frac{1 + \varphi_0^{-2}}{n},$$

using $(n+1)N = n^2 - 1 \geq n$.

(h) G_S is a principal submatrix of $G_{S'} \succ 0$, so $G_S \succ 0$ (or $S = \emptyset$). By (3), and since $(G_S + \frac{1}{n+1} I)^{-1} \preceq G_S^{-1}$, $R_2 \succeq R_0 := G_{JJ} - G_{JS} G_S^{-1} G_{SJ}$ (with $R_0 = G_{JJ}$ if $S = \emptyset$). For $x \in \mathbb{R}^j$, $x^\top R_0 x = \min_t (t, x)^\top G_{S'}(t, x) \geq N\varphi_0^2 \|x\|^2$. Hence $\det((n+1)R_2 + I) \geq ((n+1)N\varphi_0^2 + 1)^j$. Also $R_1 \preceq G_{JJ}$ and $\lambda_{\max}(G_{JJ}) \leq \mathrm{tr}\, G_{JJ} = Nj$, so $\det(R_1 + I) \leq (Nj + 1)^j \leq (N(j+1))^j$. Therefore

$$\frac{D_{S'}}{D_S} \geq \Big(\frac{(n+1)N\varphi_0^2}{N(j+1)}\Big)^j \geq \Big(\frac{n\varphi_0^2}{j+1}\Big)^j.$$

(i) If $s_0 = 0$, then $\mu = 0$, $\theta^* = 0$ and $\delta = 0$. Otherwise $\|\mu\|^2 = \theta^{*\top}_{S_0} G_{S_0} \theta^*_{S_0} \geq N\varphi_0^2 \|\theta^*\|^2$, so $\|\theta^*\|^2 \leq nM/(N\varphi_0^2)$. Hence $\delta \leq M + M/(N\varphi_0^2) \leq M(1 + \varphi_0^{-2})$. □

3 The noise event

Lemma 3.1 (Two tail bounds)**.** *(a) If $\xi \sim \chi^2_r$ with $r \geq 0$ an integer ($\chi^2_0 \equiv 0$), then $\mathbb{P}(\xi \geq 2r \log 2 + 4\tau) \leq e^{-\tau}$ for every $\tau \geq 0$. (b) If $\zeta \sim N(0, s^2)$ with $s \geq 0$, then $\mathbb{P}(|\zeta| > s\sqrt{2\tau}) \leq 2e^{-\tau}$ for every $\tau \geq 0$.*

Proof. (a) $\mathbb{E}e^{\xi/4} = 2^{r/2}$; apply Markov's inequality to $e^{\xi/4}$. (b) The case $s = 0$ is trivial. Otherwise use $\mathbb{P}(N(0,1) > x) \le e^{-x^2/2}$ for $x \ge 0$ (Chernoff) on both tails, with $x = \sqrt{2\tau}$. □

Define the events

$$\begin{aligned}
\mathcal{E}_1 &:= \{E \le 2N\},\\
\mathcal{E}_2 &:= \{\|P_S e\|^2 \le 14\sigma_\star^2 |S|\ell_n \text{ for every } S \subseteq [p]\},\\
\mathcal{E}_3 &:= \{|W_S| \le 2\sigma_\star\sqrt{a_S(|S|+1)\ell_n} \text{ for every } S \subseteq [p]\},\\
\mathcal{E}_4(u) &:= \{|E - N| \le u\sqrt{2N}\} \quad (u \ge 1),
\end{aligned}$$

and $\Omega_n := \mathcal{E}_1 \cap \mathcal{E}_2 \cap \mathcal{E}_3$. Put

$$B := 2(1+b_0)(M + 2\sigma_\star^2), \qquad c_0 := \frac{1}{2B}, \qquad \bar{b} := 8b_0(M + 2\sigma_\star^2).$$

Lemma 3.2 (The noise event)**.** $\mathbb{P}_0(\mathcal{E}_1^c) \le 2/N$, $\mathbb{P}_0(\mathcal{E}_2^c) \le 1/n$, $\mathbb{P}_0(\mathcal{E}_3^c) \le 4/n$, $\mathbb{P}_0(\mathcal{E}_4(u)^c) \le u^{-2}$, *and* $\mathbb{P}_0(\Omega_n^c) \le 8/n$. *These bounds involve neither p, nor the design, nor* θ^**, nor the cap. Under R, on* $\mathcal{E}_1$*:*

$$0 < 2b \le \bar{b} \qquad \textit{and} \qquad 2b + Q_S \le nB \quad \textit{for every } S. \tag{4}$$

Proof. $\mathcal{E}_1$ and $\mathcal{E}_4(u)$: Chebyshev, with $\mathbb{E}E = N$ and $\operatorname{Var} E = 2N$. On $\mathcal{E}_1$, $\|y_c\|^2 \le 2\|\mu\|^2 + 2\sigma_\star^2 E \le 2nM + 4n\sigma_\star^2$. So $2b = 2b_0\|y_c\|^2/N \le 2b_0(n/N)(2M + 4\sigma_\star^2) \le \bar{b}$ (as $n/N \le 2$), and, by Lemma 2.1(b), $2b + Q_S \le (1 + 2b_0/N)\|y_c\|^2 \le (1+b_0)\, n(2M + 4\sigma_\star^2) = nB$. Positivity: $y_c \ne 0$.

$\mathcal{E}_2$: For $|S| = k \ge 1$, $\|P_S e\|^2/\sigma_\star^2 \sim \chi^2_r$ with $r = \operatorname{rank} Z_S \le k$. Put $t_k := k\log p + k\log 2 + \ell_n$. Lemma 3.1(a) and $r \le k$ give $\mathbb{P}_0(\|P_S e\|^2 \ge \sigma_\star^2(2k\log 2 + 4t_k)) \le e^{-t_k} = p^{-k}2^{-k}e^{-\ell_n}$. There are $\binom{p}{k} \le p^k$ sets of size k, so the union over all $k \ge 1$ has probability at most $e^{-\ell_n}\sum_k 2^{-k} \le e^{-\ell_n} \le 1/n$. Off this union, $\|P_S e\|^2 < \sigma_\star^2(2k\log 2 + 4k\log p + 4k\log 2 + 4\ell_n) \le 14\sigma_\star^2 k\ell_n$, using $\log 2, \log p \le \ell_n$ and $k \ge 1$. For $S = \emptyset$ both sides vanish.

$\mathcal{E}_3$: $W_S \sim N(0, \sigma_\star^2 v_S)$ with $v_S \le a_S$ (Lemma 2.1(b)). With t_k as above for $k \ge 0$, Lemma 3.1(b) and the union over all sets give probability at most $\sum_{k\ge 0} p^k \cdot 2p^{-k}2^{-k}e^{-\ell_n} = 4e^{-\ell_n} \le 4/n$. Off this union, since $t_k \le (2k+1)\ell_n \le 2(k+1)\ell_n$, $|W_S| \le \sigma_\star\sqrt{2a_S \cdot 2(k+1)\ell_n}$.

Finally $2/N \le 3/n$ for $n \ge 4$, so $\mathbb{P}_0(\Omega_n^c) \le 8/n$. □

4 Models with a large approximation error lose

For an integer $d \ge 0$ put

$$C_1 := 22\sigma_\star^2 + 2\bar{\delta}, \qquad \rho(d) := \frac{2}{c_0}\big[3(s_0+1)+1\big]\ell_n + 2C_1(d + s_0 + 1)\ell_n, \qquad L(d) := (d + s_0 + 1)\ell_n,$$

$$C_a := \frac{8}{c_0} + 2C_1, \qquad \mathcal{B}(d) := \{S \in \Omega : |S| \le d,\ a_S > \rho(d)\}.$$

Then $\rho(d) \le C_a L(d)$, because $3(s_0+1)+1 \le 4(s_0+1) \le 4(d+s_0+1)$.

Proposition 4.1 (Bad models lose)**.** *Assume R and* $s_0 \le K$*. On* Ω_n*, for every integer* $d \ge 0$*,* $\Pi(\mathcal{B}(d) \mid y) \le e^{-\ell_n} \le 1/n$.

Proof. Fix $S \in \mathcal{B}(d)$ with $k = |S| \le d$, and put $\Delta_S := Q_S - Q_{S_0}$. By Lemma 2.1(c),

$$\Delta_S = a_S - a_{S_0} + 2W_S - 2W_{S_0} - e^\top M_S e + e^\top M_{S_0} e \ge a_S - a_{S_0} - 2|W_S| - 2|W_{S_0}| - \|P_S e\|^2.$$

On $\mathcal{E}_3$, $2|W_S| \le 4\sigma_\star\sqrt{a_S(k+1)\ell_n} \le \frac{a_S}{2} + 8\sigma_\star^2(k+1)\ell_n$, since $4\sqrt{xy} \le \frac{x}{2} + 8y$. Since $a_{S_0} \le \delta \le \bar\delta$ (Lemma 2.1(b),(i)), $2|W_{S_0}| \le 4\sigma_\star\sqrt{\bar\delta(s_0+1)\ell_n} \le \bar\delta + 4\sigma_\star^2(s_0+1)\ell_n$. On $\mathcal{E}_2$, $\|P_S e\|^2 \le 14\sigma_\star^2 k\ell_n$. Hence

$$\Delta_S \ge \frac{a_S}{2} - 2\bar\delta - \sigma_\star^2\big[8(k+1) + 4(s_0+1) + 14k\big]\ell_n \ge \frac{a_S}{2} - C_1(k+s_0+1)\ell_n, \tag{5}$$

because $22k + 4s_0 + 12 \le 22(k+s_0+1)$ and $2\bar\delta \le 2\bar\delta(k+s_0+1)\ell_n$. Since $k \le d$ and $a_S > \rho(d) \ge 2C_1(d+s_0+1)\ell_n$, (5) gives $\Delta_S > 0$.

By (2), Lemma 2.1(d) ($D_S \ge 1$, $D_{S_0} \le (n+1)^{s_0}$) and $\log(1+x) \ge x/(1+x)$ for $x \ge 0$,

$$\log\frac{\pi(S)}{\pi(S_0)} \le \log\frac{\varpi(S)}{\varpi(S_0)} + \frac{s_0}{2}\log(n+1) - \nu\frac{\Delta_S}{2b+Q_S} \le \log\frac{\varpi(S)}{\varpi(S_0)} + s_0\ell_n - c_0\Delta_S.$$

The last step uses $\nu \ge n/2$ and $2b + Q_S \le nB$ from (4). By Lemma 1.3(c), $-\log\varpi(S_0) \le \log 2 + s_0\log(2p) \le (2s_0+1)\ell_n$. By (5), $k \le d$, $a_S > \rho(d)$ and the definition of $\rho(d)$,

$$c_0\Delta_S > \frac{c_0\rho(d)}{2} - c_0C_1(d+s_0+1)\ell_n = \big[3(s_0+1)+1\big]\ell_n.$$

Hence $\pi(S)/\pi(S_0) < \varpi(S)e^{-\ell_n}$. (The same estimates give $e^{-3\ell_n}$; $e^{-\ell_n}$ is all that is needed.) Summing over $\mathcal{B}(d)$ and using $\pi(S_0) \le 1$ and $\sum_\Omega \varpi = 1$ (Lemma 1.3(a)) gives the claim. □

5 The master theorem

Put $\mathcal{G}(d) := \{S \in \Omega : |S| \le d,\ a_S \le \rho(d)\}$ and $r_n(d,u) := L(d)/n + u/\sqrt{n}$, and

$$C_b := 2C_a + 28\sigma_\star^2, \quad C_f := 2C_b + 2, \quad C_c := 3C_a + 16\sigma_\star^2,$$

$$C_d := \frac{\bar b + 1.02\sigma_\star^2 + C_c}{\sigma_\star^2} + \sqrt{2}, \quad C_e := 2C_d + 2\sqrt{2}.$$

These constants depend only on $(\sigma_\star^2, M, \varphi_0)$ (and $b_0 = 0.01$).

Theorem 5.1 (Master theorem)**.** *Assume R and $s_0 \le K$. For every integer $d \ge 0$:*

(i) on Ω_n, $\Pi^\dagger\big(\|X\theta - \mu\|^2 > C_f L(d) \mid y\big) \le \Pi(|S| > d \mid y) + e^{-\ell_n} + 2B/\ell_n$;

(ii) if $n \ge 8$, then for every $u \in [1, \sqrt{n/8}]$, on $\Omega_n \cap \mathcal{E}_4(u)$, $\Pi^\dagger\big(|v/\sigma_\star^2 - 1| > C_e r_n(d,u) \mid y\big) \le \Pi(|S| > d \mid y) + e^{-\ell_n} + u^{-2}$.

Proof. By Proposition 4.1, on Ω_n,

$$\Pi(\Omega \setminus \mathcal{G}(d) \mid y) \le \Pi(|S| > d \mid y) + \Pi(\mathcal{B}(d) \mid y) \le \Pi(|S| > d \mid y) + e^{-\ell_n}. \tag{6}$$

Fix $S \in \mathcal{G}(d)$, $k = |S| \le d$. It suffices to bound the conditional probabilities given (S, y) by $2B/\ell_n$ in (i) and by u^{-2} in (ii), then average over $S \sim \Pi(\cdot \mid y)$ restricted to $\mathcal{G}(d)$ and add (6).

(i) By (1), $\|X\theta - \mu\| = \|Z_S\theta_S - \mu_c\|$. By Lemma 2.1(e), $\mathcal{E}_2$ and $a_S \le \rho(d) \le C_aL(d)$, we get $\|M_S y_c - \mu_c\|^2 \le 2C_aL(d) + 28\sigma_\star^2 k\ell_n \le C_bL(d)$. Hence $\|Z_S\theta_S - \mu_c\|^2 \le 2\|Z_S\theta_S - M_S y_c\|^2 + 2C_bL(d)$, and $\{\|X\theta - \mu\|^2 > C_fL(d)\} \subseteq \{\|Z_S\theta_S - M_S y_c\|^2 > L(d)\}$. By Lemma 1.1(c), given (v, S, y),

$Z_S\theta_S - M_S y_c \sim N_N(0, vM_S)$, with second moment $v \operatorname{tr} M_S \le vk$. By Lemma 1.1(b), $\mathbb{E}[v \mid S, y] = (2b + Q_S)/(2\nu - 2) = (2b + Q_S)/(n - 1.98)$. By Markov's inequality and (4),

$$\Pi^\dagger\big(\|Z_S\theta_S - M_S y_c\|^2 > L(d) \mid S, y\big) \le \frac{k}{(d + s_0 + 1)\ell_n} \cdot \frac{nB}{n - 1.98} \le \frac{2B}{\ell_n},$$

since $n/(n - 1.98) \le 2$ for $n \ge 4$.

(ii) By Lemma 2.1(c), $\mathcal{E}_2$, $\mathcal{E}_3$, $a_S \le \rho(d)$ and $4\sqrt{xy} \le 2x + 2y$,

$$|Q_S - \sigma_\star^2 E| \le \rho(d) + 4\sigma_\star\sqrt{\rho(d)(k+1)\ell_n} + 14\sigma_\star^2 k\ell_n \le 3\rho(d) + 16\sigma_\star^2(k+1)\ell_n \le C_c L(d).$$

Put $x_S := (2b + Q_S)/(2\nu\sigma_\star^2)$. Since $2\nu = n + 0.02 = N + 1.02$,

$$2b + Q_S - 2\nu\sigma_\star^2 = 2b + (Q_S - \sigma_\star^2 E) + \sigma_\star^2(E - N) - 1.02\sigma_\star^2.$$

On $\mathcal{E}_1 \cap \mathcal{E}_4(u)$, using (4), $2\nu \ge n$ and $\sqrt{2N}/n \le \sqrt{2/n}$,

$$|x_S - 1| \le \frac{\bar{b} + 1.02\sigma_\star^2}{n\sigma_\star^2} + \frac{C_c L(d)}{n\sigma_\star^2} + \frac{\sqrt{2}\,u}{\sqrt{n}} \le C_d\, r_n(d, u),$$

since $1/n \le u/\sqrt{n} \le r_n(d, u)$. By Lemma 1.1(b), $\Upsilon := (2b + Q_S)/(2v) \sim \text{Gamma}(\nu, 1)$ given (S, y), and $v/\sigma_\star^2 = x_S\,\nu/\Upsilon$. By Chebyshev, $\Pi^\dagger(|\Upsilon - \nu| > u\sqrt{\nu} \mid S, y) \le u^{-2}$. On $\{|\Upsilon - \nu| \le u\sqrt{\nu}\}$, $u \le \sqrt{n/8} \le \sqrt{\nu}/2$ gives $\Upsilon \ge \nu/2$, so $|\nu/\Upsilon - 1| \le 2u/\sqrt{\nu} \le 2\sqrt{2}u/\sqrt{n} \le 1$. For reals x, g with $|g-1| \le 1$, $|xg-1| \le |x-1|g + |g-1| \le 2|x-1| + |g-1|$. Hence $|v/\sigma_\star^2 - 1| \le 2C_d r_n(d, u) + 2\sqrt{2}u/\sqrt{n} \le C_e r_n(d, u)$ there. □

6 Part (i) and part (iii) under G, R

Lemma 6.1. $\log_2 n \le m_n \le 5(s_0 + 1)\ell_n$. *Hence* $L(m_n) \le 6(s_0 + 1)\ell_n^2 = 6n\varepsilon_{2,n}^2$.

Proof. The numerator in m_n is at least $\log n$ (from the term $\frac{s_0+1}{2}\log n + \frac{1}{2}\log n \ge \log n$), which gives the lower bound. For the upper bound, it is at most $s_0(\log 2 + \log p) + \frac{s_0+2}{2}\log n \le (2s_0 + \frac{s_0+2}{2})\ell_n \le 2.5(s_0 + 1)\ell_n$. Since $1/\log 2 < 1.443$, $m_n \le 3.61(s_0 + 1)\ell_n + 1 \le 5(s_0 + 1)\ell_n$. Then $L(m_n) = (m_n + s_0 + 1)\ell_n \le 6(s_0 + 1)\ell_n^2$, as $s_0 + 1 \le (s_0 + 1)\ell_n$. □

Proof of (i). Let n be large enough that $s_0 \le K$ (R). Theorem 5.1(i) with $d = m_n$ and Lemma 6.1 give, on Ω_n,

$$\Pi^\dagger\big(\|X\theta - \mu\|^2/n > 6C_f\varepsilon_{2,n}^2 \mid y\big) \le \Pi(|S| > m_n \mid y) + e^{-\ell_n} + 2B/\ell_n.$$

Fix $\epsilon_0 > 0$. Since $\ell_n \ge \log n \to \infty$, for all large n we have $e^{-\ell_n} + 2B/\ell_n \le \epsilon_0/2$. Then

$$\mathbb{P}_0\Big(\Pi^\dagger\big(\|X\theta - \mu\|^2/n > 6C_f\varepsilon_{2,n}^2 \mid y\big) > \epsilon_0\Big) \le \frac{8}{n} + \mathbb{P}_0\big(\Pi(|S| > m_n \mid y) > \epsilon_0/2\big) \to 0,$$

by Lemma 3.2 and H-T02. □

Proof of (iii) under G, R. Let $u_n \to \infty$ be deterministic, and put $u_n' := \min\{u_n, \sqrt{n/8}\}$. Then $u_n' \to \infty$, $u_n' \le u_n$, and $u_n' \ge 1$ for all large n. By Lemma 6.1, $r_n(m_n, u_n') \le 6\varepsilon_{2,n}^2 + u_n'/\sqrt{n} \le 6(\varepsilon_{2,n}^2 + u_n/\sqrt{n})$. Theorem 5.1(ii) with $d = m_n$ and $u = u_n'$ gives, on $\Omega_n \cap \mathcal{E}_4(u_n')$,

$$\Pi^\dagger\big(|v/\sigma_\star^2 - 1| > 6C_e[\varepsilon_{2,n}^2 + u_n/\sqrt{n}] \mid y\big) \le \Pi(|S| > m_n \mid y) + e^{-\ell_n} + (u_n')^{-2}.$$

As in the proof of (i), the $\mathbb{P}_0$-probability that the left side exceeds ϵ_0 is at most $8/n + (u_n')^{-2} + \mathbb{P}_0(\Pi(|S| > m_n \mid y) > \epsilon_0/2) \to 0$, for all n large enough that $e^{-\ell_n} + (u_n')^{-2} \le \epsilon_0/2$. □

7 The sharp dimension bound under SE

Throughout this section, assume G, R, SE and H-T02, and put

$$j_0 := \lceil s_0(2A+1) \rceil, \qquad d_* = s_0 + j_0, \qquad u_n^\sharp := \log n.$$

Let n_1 be such that, for all $n \ge n_1$: $s_0 \le K$; $m_n + s_0 \le \lfloor n/2 \rfloor$; and $d_* < m_n$. Such an n_1 exists by R, because $m_n = O(\log n)$ and $m_n \ge \log_2 n$ (Lemma 6.1). For $n \ge n_1$:

$$K = \min(p, \lfloor n/2 \rfloor) \ge \min(p, m_n + s_0). \tag{7}$$

So every set T with $|T| \le m_n + s_0$ lies in Ω.

Lemma 7.1 (An underfitting model is dominated by its union with S_0)**.** *Let $n \ge n_1$, $S \in \Omega$ with $|S| \le m_n$, and $S' := S \cup S_0$. Then $S' \in \Omega$ and*

$$\frac{\pi(S)}{\pi(S_0)} \le (2p)^{s_0}(n+1)^{s_0/2} \frac{\pi(S')}{\pi(S_0)}.$$

Proof. $|S'| \le \min(p, m_n + s_0) \le K$ by (7). By Lemma 1.3(b), $\varpi(S)/\varpi(S') = w(|S|)/w(|S'|) \le (2p)^{|S'|-|S|} \le (2p)^{s_0}$. By (2) with Lemma 2.1(f) ($Q_{S'} \le Q_S$, $D_{S'}/D_S \le (n+1)^{|S_0 \setminus S|}$), $\pi(S)/\pi(S') \le (2p)^{s_0}(n+1)^{s_0/2}$. □

Lemma 7.2 (A truncated exponential moment)**.** *Let $\xi \sim \chi^2_j$ with $j \ge 1$, $\eta \ge 0$ and $\tau > 0$. Then $\mathbb{E}\big[e^{(1+\eta)\xi/2}\mathbf{1}\{\xi \le \tau\}\big] \le (e\tau/j)^{j/2} e^{\eta\tau/2}$.*

Proof. With the χ^2_j density, the expectation equals

$$\frac{1}{2^{j/2}\Gamma(j/2)} \int_0^\tau x^{j/2-1} e^{\eta x/2} dx \le \frac{e^{\eta\tau/2}\tau^{j/2}}{2^{j/2}\Gamma(j/2)(j/2)} = \frac{(\tau/2)^{j/2} e^{\eta\tau/2}}{\Gamma(j/2+1)}.$$

Moreover $\Gamma(x+1) \ge \int_x^\infty t^x e^{-t} dt \ge x^x e^{-x}$ for $x > 0$, so $(\tau/2)^{j/2}/\Gamma(j/2+1) \le (e\tau/j)^{j/2}$. □

Define the deterministic sequences

$$\vartheta_n := \frac{\sqrt{2}\, u_n^\sharp}{\sqrt{N}} + \frac{4\sqrt{\bar\delta(m_n + s_0 + 1)\ell_n}}{\sigma_\star N} + \frac{14(m_n + s_0)\ell_n}{N}, \qquad \eta_n := \frac{n + 0.02}{N(1 - \vartheta_n)} - 1 \quad (\vartheta_n < 1),$$

and, for $j \ge 1$,

$$\zeta_j := \bar\delta + 8\sigma_\star\sqrt{\bar\delta(s_0 + j + 1)\ell_n} + \frac{14(1 + \varphi_0^{-2})\sigma_\star^2 s_0 \ell_n}{n}, \qquad \tau_j := 14(s_0 + j)\ell_n.$$

By Lemma 6.1 and $\ell_n \le A \log n$, $(m_n + s_0 + 1)\ell_n = O(\log^2 n)$, so $\vartheta_n = O(\log n/\sqrt{n}) + O(\log n/n) + O(\log^2 n/n)$ and $\vartheta_n \ell_n \to 0$. When $\vartheta_n \le \frac{1}{2}$, $\eta_n = \frac{1.02 + N\vartheta_n}{N(1-\vartheta_n)} \le 2\vartheta_n + \frac{2.04}{N}$, so $\eta_n \ge 0$ and $\eta_n \ell_n \to 0$. Let $n_2 \ge n_1$ be such that $\vartheta_n \le \frac{1}{2}$ for $n \ge n_2$.

For $J \subseteq [p] \setminus S_0$, $|J| = j$, put $S' = S_0 \cup J$,

$$\chi^2_J := \|(P_{S'} - P_{S_0})e\|^2/\sigma_\star^2, \qquad \tilde T_j := \sum_{J \subseteq [p]\setminus S_0,\ |J| = j} e^{(1+\eta_n)\chi^2_J/2}\, \mathbf{1}\{\chi^2_J \le \tau_j\}.$$

Proposition 7.3 (Supersets of S_0)**.** *Let $n \ge n_2$ and $1 \le j \le m_n$. On $\mathcal{E}_2 \cap \mathcal{E}_3 \cap \mathcal{E}_4(u_n^\sharp)$,*

$$\sum_{J \subseteq [p]\setminus S_0,\ |J|=j} \frac{\pi(S_0 \cup J)}{\pi(S_0)} \le \Phi_j \tilde{T}_j, \qquad \Phi_j := \Big(\frac{s_0+j}{p}\Big)^j \Big(\frac{j+1}{n\varphi_0^2}\Big)^{j/2} \exp\Big(\frac{1+\eta_n}{2\sigma_\star^2}\zeta_j\Big), \tag{8}$$

and

$$\mathbb{E}_0\big[\Phi_j \tilde{T}_j\big] \le \Big(\frac{28e^3(s_0+1)^2(s_0+j)\ell_n}{n\varphi_0^2}\Big)^{j/2} \exp\Big(7\eta_n(s_0+j)\ell_n + \frac{1+\eta_n}{2\sigma_\star^2}\zeta_j\Big). \tag{9}$$

(If $s_0 + j > p$ the sum in (8) and $\tilde{T}_j$ are empty, and both bounds hold trivially.)

Proof. Fix J with $|J| = j$ and $S' = S_0 \cup J$. Then $1 \le |S'| = s_0 + j \le \min(p, m_n + s_0)$, so $S' \in \Omega$ by (7). By SE, $\lambda_{\min}(G_{S'}) \ge N\varphi_0^2$. In particular $Z_{S'}$ has full column rank, $P_{S'} - P_{S_0}$ is an orthogonal projection of rank j, and $\chi_J^2 \sim \chi_j^2$ under $\mathbb{P}_0$. If $s_0 \ge 1$, R gives $\lambda_{\min}(G_{S_0}) \ge N\varphi_0^2$. If $s_0 = 0$, $M_{S_0} = P_{S_0} = 0$ and the terms involving S_0 below vanish.

Prior. By Lemma 1.3(d), $\varpi(S')/\varpi(S_0) = w(s_0+j)/w(s_0) \le ((s_0+j)/p)^j$.

Occam factor. By Lemma 2.1(h) with $S = S_0$ and $\|z_i\|^2 = N$: $(D_{S'}/D_{S_0})^{-1/2} \le ((j+1)/(n\varphi_0^2))^{j/2}$.

Fit. By Lemma 2.1(c) for S_0 and S',

$$Q_{S_0} - Q_{S'} = (a_{S_0} - a_{S'}) + 2(W_{S_0} - W_{S'}) + e^\top (M_{S'} - M_{S_0})e.$$

Here $a_{S_0} - a_{S'} \le a_{S_0} \le \bar{\delta}$. On $\mathcal{E}_3$, with $a_{S_0}, a_{S'} \le \bar{\delta}$ (Lemma 2.1(b),(i), since $S' \supseteq S_0$), $2|W_{S_0}| + 2|W_{S'}| \le 8\sigma_\star\sqrt{\bar{\delta}(s_0+j+1)\ell_n}$. By Lemma 2.1(a),(g), $M_{S'} \preceq P_{S'}$ and $P_{S_0} - M_{S_0} \preceq \frac{1+\varphi_0^{-2}}{n} P_{S_0}$. Hence, on $\mathcal{E}_2$,

$$e^\top (M_{S'} - M_{S_0})e \le e^\top (P_{S'} - P_{S_0})e + e^\top (P_{S_0} - M_{S_0})e \le \sigma_\star^2 \chi_J^2 + \frac{1+\varphi_0^{-2}}{n}\, 14\sigma_\star^2 s_0 \ell_n.$$

So $Q_{S_0} - Q_{S'} \le \sigma_\star^2 \chi_J^2 + \zeta_j$. For the denominator, by Lemma 2.1(c),(a), $b > 0$, $\mathcal{E}_2 \cap \mathcal{E}_3 \cap \mathcal{E}_4(u_n^\sharp)$, $a_{S'} \le \bar{\delta}$ and $s_0 + j \le m_n + s_0$,

$$\begin{aligned} 2b + Q_{S'} &\ge \sigma_\star^2 E - 2|W_{S'}| - \|P_{S'} e\|^2 \\ &\ge \sigma_\star^2 (N - u_n^\sharp \sqrt{2N}) - 4\sigma_\star \sqrt{\bar{\delta}(m_n + s_0 + 1)\ell_n} - 14\sigma_\star^2 (m_n + s_0)\ell_n = N\sigma_\star^2(1 - \vartheta_n) > 0. \end{aligned}$$

Hence $\nu/(2b + Q_{S'}) \le (n + 0.02)/(2N\sigma_\star^2(1-\vartheta_n)) = (1+\eta_n)/(2\sigma_\star^2)$. Using $\log(1+x) \le x$ $(x > -1)$ and the fact that the bound $\sigma_\star^2\chi_J^2 + \zeta_j$ is ≥ 0,

$$\nu \log \frac{2b + Q_{S_0}}{2b + Q_{S'}} \le \frac{\nu(Q_{S_0} - Q_{S'})}{2b + Q_{S'}} \le \frac{1+\eta_n}{2\sigma_\star^2}\big(\sigma_\star^2 \chi_J^2 + \zeta_j\big).$$

Assembly. By (2), $\pi(S')/\pi(S_0) \le \Phi_j e^{(1+\eta_n)\chi_J^2/2}$. On $\mathcal{E}_2$, $\sigma_\star^2\chi_J^2 \le \|P_{S'}e\|^2 \le 14\sigma_\star^2(s_0+j)\ell_n$, since the range of $P_{S'} - P_{S_0}$ is contained in that of $P_{S'}$. So $\chi_J^2 \le \tau_j$, and the indicator in $\tilde{T}_j$ equals 1. Summing over J gives (8).

Expectation. Φ_j is deterministic. There are $\binom{p-s_0}{j} \le p^j/j!$ sets J. For each, Lemma 7.2 gives $\mathbb{E}_0[e^{(1+\eta_n)\chi_J^2/2}\mathbf{1}\{\chi_J^2 \le \tau_j\}] \le (e\tau_j/j)^{j/2} e^{\eta_n \tau_j/2}$. Hence

$$\mathbb{E}_0[\Phi_j \tilde{T}_j] \le \frac{(s_0+j)^j}{j!}\Big(\frac{j+1}{j} \cdot \frac{14e(s_0+j)\ell_n}{n\varphi_0^2}\Big)^{j/2} e^{7\eta_n(s_0+j)\ell_n + \frac{1+\eta_n}{2\sigma_\star^2}\zeta_j}.$$

Finally $(s_0+j)^j/j! \le (e(s_0+j)/j)^j \le (e(s_0+1))^j = (e^2(s_0+1)^2)^{j/2}$, using $j! \ge (j/e)^j$, and $(j+1)/j \le 2$. □

Theorem 7.4 (Sharp dimension bound)**.** *Under G, R, SE and H-T02, in R1,* $\Pi(|S| > d_* \mid y) \to 0$ *in* $\mathbb{P}_0$*-probability.*

Proof. Let $n \ge n_2$ and $\mathcal{M} := \{S \in \Omega : d_* < |S| \le m_n\}$. Then

$$\Pi(|S| > d_* \mid y) \le \Pi(|S| > m_n \mid y) + \Pi(\mathcal{M} \mid y), \tag{10}$$

and the first term tends to 0 in probability by H-T02.

Step 1 (reduction to supersets). For $S \in \mathcal{M}$ let $J := S \setminus S_0$ and $S' = S \cup S_0 = S_0 \cup J$. Then $j := |J| \ge |S| - s_0 > j_0$ and $j \le |S| \le m_n$. The map $S \mapsto S'$ is at most 2^{s_0}-to-one, since S is determined by S' and $S \cap S_0$. By Lemma 7.1 and $\pi(S_0) \le 1$,

$$\Pi(\mathcal{M} \mid y) \le \sum_{S \in \mathcal{M}} \frac{\pi(S)}{\pi(S_0)} \le (2p)^{s_0}(n+1)^{s_0/2} 2^{s_0} \sum_{j=j_0+1}^{m_n} \sum_{J \subseteq [p] \setminus S_0,\ |J|=j} \frac{\pi(S_0 \cup J)}{\pi(S_0)}.$$

Step 2 (the good event). By Proposition 7.3, on $\mathcal{E}_2 \cap \mathcal{E}_3 \cap \mathcal{E}_4(u_n^\sharp)$,

$$\Pi(\mathcal{M} \mid y) \le \Xi_n := (2p)^{s_0}(n+1)^{s_0/2} 2^{s_0} \sum_{j=j_0+1}^{m_n} \Phi_j \tilde{T}_j.$$

Ξ_n is a nonnegative random variable defined for every e.

Step 3 (the expectation of Ξ_n*).* For $1 \le j \le m_n$, $s_0 + j \le s_0 + m_n \le 6(s_0+1)\ell_n$ (Lemma 6.1). So (9) gives $\mathbb{E}_0[\Phi_j \tilde{T}_j] \le \kappa_n^{j/2} e^{\psi_j}$, with

$$\kappa_n := \frac{168 e^3 (s_0+1)^3 \ell_n^2}{n \varphi_0^2}, \qquad \psi_j := 7\eta_n(s_0+j)\ell_n + \frac{1+\eta_n}{2\sigma_\star^2}\zeta_j.$$

For $j \ge 1$: $s_0 + j \le (s_0+1)j$, $s_0 + j + 1 \le 2(s_0+1)j$ and $\sqrt{j} \le j$. Hence $\psi_j \le (\alpha_n + \beta_n)j + \gamma_n$, with

$$\alpha_n := 7(s_0+1)\eta_n \ell_n, \qquad \beta_n := \frac{4(1+\eta_n)}{\sigma_\star}\sqrt{2\bar{\delta}(s_0+1)\ell_n}, \qquad \gamma_n := \frac{1+\eta_n}{2\sigma_\star^2}\Big(\bar{\delta} + \frac{14(1+\varphi_0^{-2})\sigma_\star^2 s_0 \ell_n}{n}\Big).$$

By the remarks after the definition of η_n: $\alpha_n \to 0$, $\beta_n = O(\sqrt{\log n})$ and $\gamma_n = O(1)$. Put $\kappa_n' := \kappa_n e^{2\alpha_n + 2\beta_n}$. Then

$$\log \kappa_n' = -\log n + \varrho_n, \qquad \varrho_n := 2\log \ell_n + \log \frac{168 e^3 (s_0+1)^3}{\varphi_0^2} + 2\alpha_n + 2\beta_n = O(\sqrt{\log n}),$$

since $\log \ell_n \le \log(A \log n)$. In particular $\kappa_n' \le \frac{1}{4}$ for all large n, and then $\sum_{j > j_0} (\kappa_n')^{j/2} \le (\kappa_n')^{(j_0+1)/2}/(1 - \sqrt{\kappa_n'}) \le 2(\kappa_n')^{(j_0+1)/2}$. Hence the expectation $\bar{\Xi}_n := \mathbb{E}_0 \Xi_n$ satisfies

$$\bar{\Xi}_n \le (2p)^{s_0}(n+1)^{s_0/2} 2^{s_0} \cdot 2 e^{\gamma_n} (\kappa_n')^{(j_0+1)/2}.$$

Using $p \le n^A$ and $\log(n+1) \le \log n + 1$ (if $\bar{\Xi}_n = 0$ there is nothing to prove),

$$\log \bar{\Xi}_n \le \Big(s_0 A + \frac{s_0}{2} - \frac{j_0+1}{2}\Big)\log n + \frac{j_0+1}{2}\varrho_n + \gamma_n + (2s_0+1)\log 2 + \frac{s_0}{2}.$$

The coefficient of $\log n$ is $-\frac{1}{2}\big[(j_0+1) - s_0(2A+1)\big] \le -\frac{1}{2}$, because $j_0 \ge s_0(2A+1)$. The remaining terms are $O(\sqrt{\log n})$. Hence $\log \bar{\Xi}_n \to -\infty$, that is, $\bar{\Xi}_n \to 0$.

Step 4 (conclusion). By Markov's inequality, $\mathbb{P}_0(\Xi_n > \sqrt{\bar{\Xi}_n}) \le \sqrt{\bar{\Xi}_n}$. So, by Lemma 3.2, outside an event of probability at most $\frac{1}{n} + \frac{4}{n} + (\log n)^{-2} + \sqrt{\bar{\Xi}_n} \to 0$, we have $\Pi(\mathcal{M} \mid y) \le \sqrt{\bar{\Xi}_n} \to 0$. With (10) this proves the theorem. □

8 Parts (ii), (iii) under SE, and (iv)

Since $d_* = s_0 + \lceil s_0(2A+1) \rceil \le (2A+2)s_0 + 1$,

$$L(d_*) = (d_* + s_0 + 1)\ell_n \le (2A+3)(s_0+1)\ell_n = (2A+3)\, n\varepsilon_{1,n}^2. \tag{11}$$

Proof of (ii). The first claim is Theorem 7.4.

Prediction. By Theorem 5.1(i) with $d = d_*$ and (11), on Ω_n (for $n \ge n_1$),

$$\Pi^\dagger\big(\|X\theta - \mu\|^2/n > (2A+3)C_f\varepsilon_{1,n}^2 \mid y\big) \le \Pi(|S| > d_* \mid y) + e^{-\ell_n} + 2B/\ell_n.$$

As in the proof of (i), with Theorem 7.4 in place of H-T02, the left side tends to 0 in probability.

Coefficients. Let $n \ge n_1$, so that $d_* + s_0 < m_n + s_0$. Take (S, θ) with $|S| \le d_*$ and θ supported in S, and put $T := S \cup S_0$. If $T = \emptyset$, then $\theta = \theta^* = 0$. Otherwise $1 \le |T| \le \min(p, m_n + s_0)$, so SE gives $\|X(\theta - \theta^*)\|^2 = (\theta - \theta^*)_T^\top G_T (\theta - \theta^*)_T \ge N\varphi_0^2 \|\theta - \theta^*\|^2$. Since $X\theta^* = \mu$ and $n/N \le 2$,

$$\|\theta - \theta^*\|^2 \le \frac{\|X\theta - \mu\|^2}{N\varphi_0^2} \le \frac{2}{\varphi_0^2} \cdot \frac{\|X\theta - \mu\|^2}{n}.$$

Hence

$$\Big\{\|\theta - \theta^*\|^2 > \frac{2(2A+3)C_f\varepsilon_{1,n}^2}{\varphi_0^2}\Big\} \subseteq \{|S| > d_*\} \cup \Big\{\frac{\|X\theta - \mu\|^2}{n} > (2A+3)C_f\varepsilon_{1,n}^2\Big\},$$

and the $\Pi^\dagger$-probability of the right side tends to 0 in probability by the first two claims. □

Proof of (iii) under SE. With $u_n' = \min\{u_n, \sqrt{n/8}\}$ as before, (11) gives $r_n(d_*, u_n') \le (2A+3)(\varepsilon_{1,n}^2 + u_n/\sqrt{n})$. Theorem 5.1(ii) with $d = d_*$, $u = u_n'$ gives, on $\Omega_n \cap \mathcal{E}_4(u_n')$,

$$\Pi^\dagger\big(|v/\sigma_\star^2 - 1| > (2A+3)C_e[\varepsilon_{1,n}^2 + u_n/\sqrt{n}] \mid y\big) \le \Pi(|S| > d_* \mid y) + e^{-\ell_n} + (u_n')^{-2},$$

and the conclusion follows as before, with Theorem 7.4 in place of H-T02. □

Proof of (iv). D_x^{-1} is diagonal with entries at most $1/d_-$, so $\|D_x^{-1}(\theta - \theta^*)\|^2 \le d_-^{-2}\|\theta - \theta^*\|^2$. Hence $\{\|D_x^{-1}(\theta - \theta^*)\|^2 > c/d_-^2\} \subseteq \{\|\theta - \theta^*\|^2 > c\}$ for every $c > 0$. Apply this with $c = 2(2A+3)C_f\varepsilon_{1,n}^2/\varphi_0^2$ and use (ii). □

The constant. All parts hold with

$$C := \max\big\{6C_f,\ 6C_e,\ 2(2A+3)C_f,\ (2A+3)C_e\big\}, \tag{12}$$

because every event above is monotone in its threshold. C depends only on $(\sigma_\star^2, M, \varphi_0, A)$ (and $b_0 = 0.01$). It depends neither on s_0 nor on the sequences p_n, u_n.

9 Remarks

Remark 9.1 (Where each hypothesis enters).

- G: through $e \sim N_N(0, \sigma_\star^2 I)$ in Lemma 3.2 and Proposition 7.3.
- R: $\|\mu\|^2/n \le M$ in (4) and $\bar{\delta}$; $\lambda_{\min}(R_{S_0}) \ge \varphi_0^2$ in Lemma 2.1(i),(g); $s_0 \le K$ for $\varpi(S_0) > 0$ (Lemma 1.3(c)). No eigenvalue condition on any other model is used in Sections 1–6.

- $p \le n^A$: in Section 7 (through $\ell_n \le A \log n$, $m_n = O(\log n)$ in the choice of n_1, and the conversion cost $(2p)^{s_0} \le 2^{s_0} n^{As_0}$), and through H-T02, which is assumed in R1 and is used in Sections 6–8. The non-asymptotic statements of Sections 1–5 hold for every p.
- SE: only in Section 7 (full rank and Occam lower bound of $S_0 \cup J$) and in the coefficient bound of (ii).
- The cap $K = \min(p, \lfloor n/2 \rfloor)$: through $s_0 \le K$, and in Section 7 through (7). So every comparison with a superset $S \cup S_0$ stays inside Ω; no augmentation beyond the cap is used.
- H-T02: only to discard $\{|S| > m_n\}$, in Section 6 and in (10).
- The data-dependent rate $b = b_0 s_y^2$ enters through Lemmas 1.1–1.2 and through (4) ($0 < 2b \le \bar{b}$ and $2b + Q_S \le nB$ on $\mathcal{E}_1$). All comparisons of Sections 4 and 7 are pointwise in y, so no change of the rate (no mixed-rate transfer) is needed.
- No beta-min condition is used. A weak true variable missed by S costs little in a_S (Section 4), and in Section 7 it is added in $S \cup S_0$ at the fixed price $(2p)^{s_0}(n+1)^{s_0/2} 2^{s_0}$, which j_0 absorbs.

Remark 9.2 (Provenance). The argument re-proves, for the v3 target, a draft argument for the v1/v2 target (a master theorem with a free dimension cut-off, and a superset-summation dimension bound). The changes for v3 are:

- the noise dimension $N = n - 1$ of the contrast coordinates;
- column norms N;
- the IG$(0.51, b_0 s_y^2)$ prior with data-dependent rate;
- the cap K;
- the dimension input H-T02 in place of the earlier one.

Every step has been rechecked against these.